

HARDWARE TEARDOWN • REVERSE ENGINEERING

# M-TECH Digital Panel Meter Teardown Analysis

Architecture, internal circuitry, and functional breakdown of an industrial DC ammeter.

— System Analysis & Circuit Mapping

# PART-1

## PHYSICAL INSPECTION – STUDYING THE PRODUCT

- PRODUCT NAME/MODEL -> **DIGITAL PANEL MATERS**
- MANUFACTURER -> **M TECH ENGINEERING CORP.**
- APPROXIMATE AGE/VERSION -> **10 YEAR OLD**
- EXTERNAL DIMENSIONS -> **48X96X110MM(HEIGHT,WIDTH,DEPTH)**
- BUTTONS/SWITCHES -> **NONE**
- LEDS/DISPLAYS -> **4 DIGITAL SEVEN SEGMENT LED DISPLAY**
- CONNECTORS -> **SOLDERING**
- POWER SOURCE -> **AC INPUT, STEP DOWN TRANSFORMER**
- PORTS -> **NONE**
- SENSORS -> **NONE**
- MECHANICAL INTERFACES -> **BLACK PLASTIC RECTANGULAR HOUSING**
- LABELS AND MARKINGS -> **DIGITAL PANEL METER, M-TECH, A**
- VISIBLE SCREWS/SEAMS -> **6 SCREWS**
- VENTILATION/OPENINGS -> **2 VENTS BOTH SIDES**

## **WHAT IS THIS PRODUCT SUPPOSED TO DO?**

THIS PRODUCT IS A DIGITAL PANEL AMMETER, DESIGNED TO CONTINUOUSLY MEASURE AND DIGITALLY DISPLAY THE ELECTRICAL CURRENT FLOWING THROUGH A CIRCUIT .

## **WHAT DOES THE USER EXPECT?**

THE USER NEEDS A RELIABLE INDUSTRIAL DC AMMETER THAT PROVIDES INSTANT, HIGH-VISIBILITY CURRENT READINGS FROM A DISTANCE. TO MAKE THIS PRODUCT.

# PART-2

## 1).DIFFERENCE IN TECHNOLOGY ACROSS VERSIONS

### EARLY STAGE:

- MOVING-COIL ANALOG METER (1820–1880).
- SIMPLE, NEEDS NO POWER SUPPLY, CHEAP.

### DRAWBACK:

NEEDLE READING ERROR, FRAGILE, AND IT LOADS THE CIRCUIT.

## 2). FIRST GENERATION:

VACUUM-TUBE VOLTMETER AND FIRST DIGITAL VOLTMETER (1930–1950).

### IMPROVEMENT:

HIGH INPUT IMPEDANCE. THE FIRST DIGITAL VOLTMETERS APPEARED IN THE 1950S.

### DRAWBACK:

BULKY, NEEDS WARM-UP, DRIFTS, EXPENSIVE.

### 3). SECOND GENERATION:

SINGLE-CHIP ADC (LATE 1970S). THIS PRODUCT  
IMPROVEMENT:

THE ICL7107 DOES THE WHOLE DUAL-SLOPE ADC, AUTO-ZERO, POLARITY DETECTION AND LED DRIVING IN ONE IC. PARTS COUNT, COST AND SIZE DROP SHARPLY.

#### DRAWBACK:

NEEDS  $\pm 5\text{ V}$  SUPPLIES, WHICH IS WHY THIS BOARD HAS A TRANSFORMER, DIODES AND A ZENER . THE LEDS DRAW HIGH CURRENT. IT READS ONLY  $3\frac{1}{2}$  DIGITS, SLOWLY (ABOUT 3 READINGS PER SECOND), AND NEEDS MANUAL CALIBRATION WITH VR1.

**LATEST GENERATION:**

**MICROCONTROLLER AND SIGMA-DELTA ADC**

**IMPROVEMENT:**

**16–24 BIT RESOLUTION, AUTO-RANGING, TRUE-RMS, A SINGLE 3.3–5 V SUPPLY, AND  
OLED, LCD OR WIRELESS OUTPUT .**

**DRAWBACK:**

**HIGHER COST, NEEDS FIRMWARE, AND SAMPLING NOISE.**

## 2). CURRENT STATUS: CORE SMART TECHNOLOGIES

- MICROCONTROLLER (ARDUINO, ESP32) WITH A SIGMA-DELTA ADC (ADS1115, INA219).
- BLUETOOTH AND WI-FI FOR IOT MONITORING AND CLOUD DATA LOGGING .
- AUTO-RANGING, DATA LOGGING, ALARMS, OLED/TFT DISPLAYS, USB-C POWER.

### 3. ENGINEERING IMPORTANCE OF THIS EVOLUTION

- ONE CHIP REPLACED DOZENS OF COMPONENTS, SO SIZE AND COST FELL.
- ACCURACY AND RESOLUTION IMPROVED, AND READING ERRORS DISAPPEARED .
- IT MADE MEASUREMENT AUTOMATIC, SO DATA COULD BE LOGGED AND MONITORED REMOTELY .
- IT IS A CORE MIXED-SIGNAL (ANALOG + DIGITAL) DESIGN EXAMPLE



## PART-III INTERNAL COMPONENTS AND WORKING FLOW

### PHYSICAL INTERFACE & NAMEPLATE

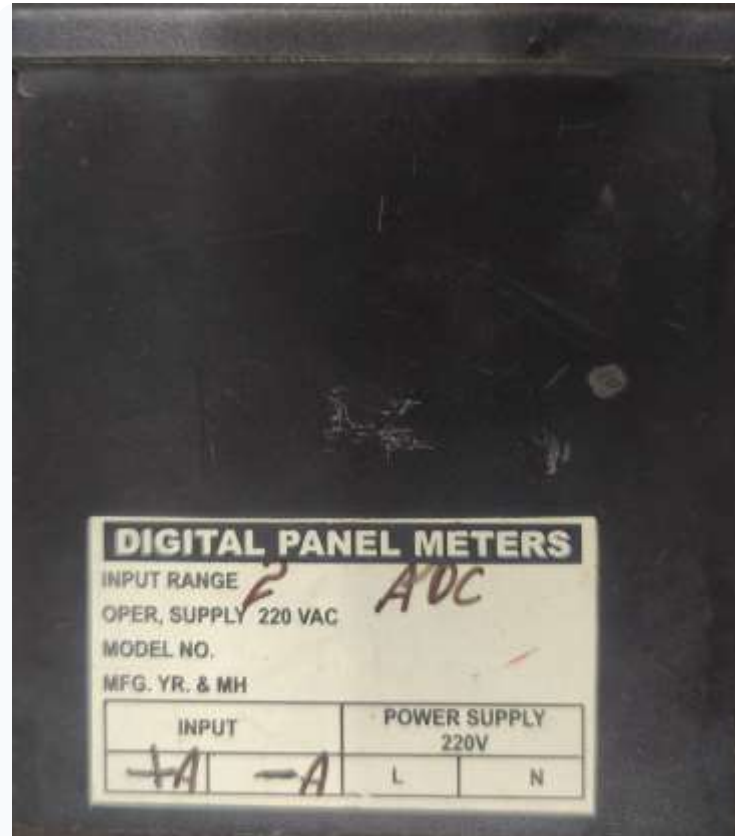
## Exterior Inspection & Rating Label



**FIG 1.0 • FRONT FASCIA**

### Bezel & Display Filter

High-contrast red acrylic lens with DC ammeter glyph.



**FIG 1.1 • SPECIFICATION PLATE**

### Rating & Terminal Pinout

Manufacturer label marking 2 ADC range and 220V supply.

### CURRENT INPUT RANGE

**0 to 2.000A DC**

Direct shunt measurement range.

### AUXILIARY POWER SUPPLY

**220V AC Mains (L / N)**

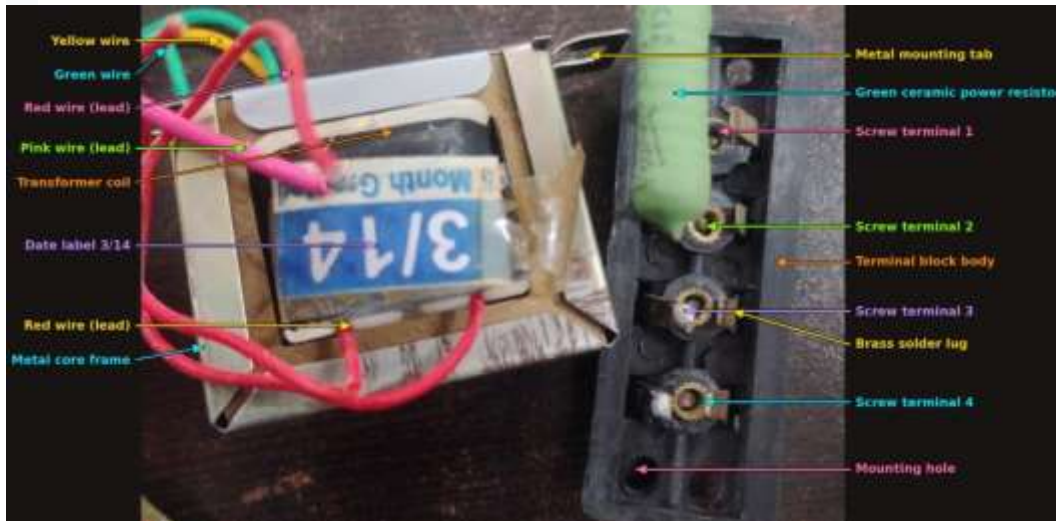
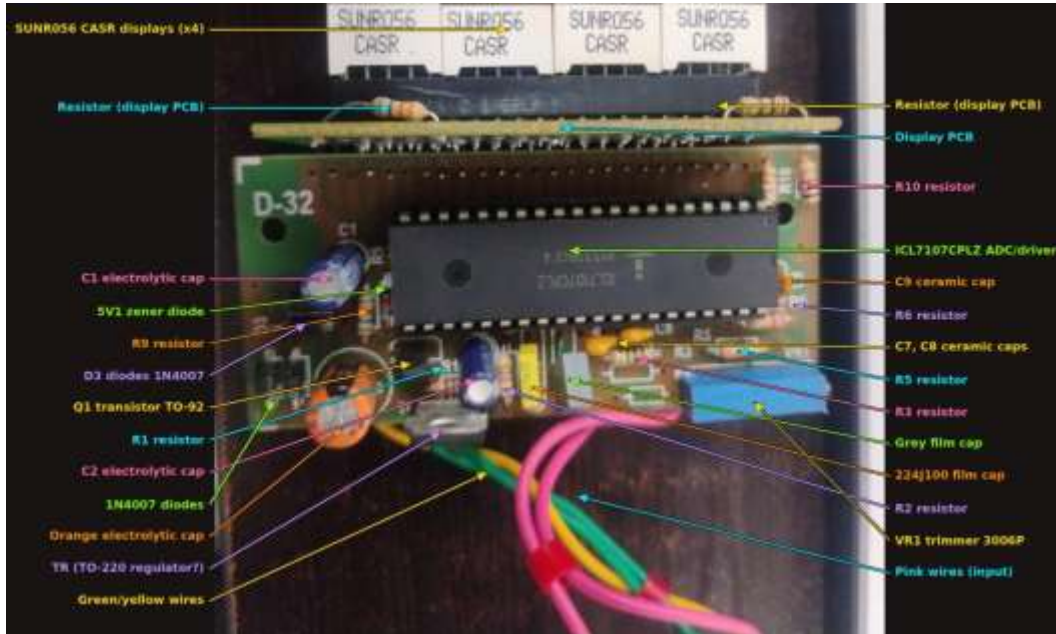
Isolated operating power rails.

### ENCLOSURE STANDARD

**DIN 48 × 96 mm**

Standard industrial panel mount.

# Internal Hardware Subsystems & Labels



IC1

## ICL7107 3.5-Digit ADC

DIP-40 dual-slope converter with built-in LED segment drivers.

DISP

## SUNR056 7-Segment Daughterboard

Four common-anode red LED numeric display modules.

XFRM

## Mains Step-Down Transformer

220V iron-core transformer providing galvanic isolation.

SHUNT

## Current Shunt Resistor

Precision green wirewound shunt mounted across input posts.

VR1

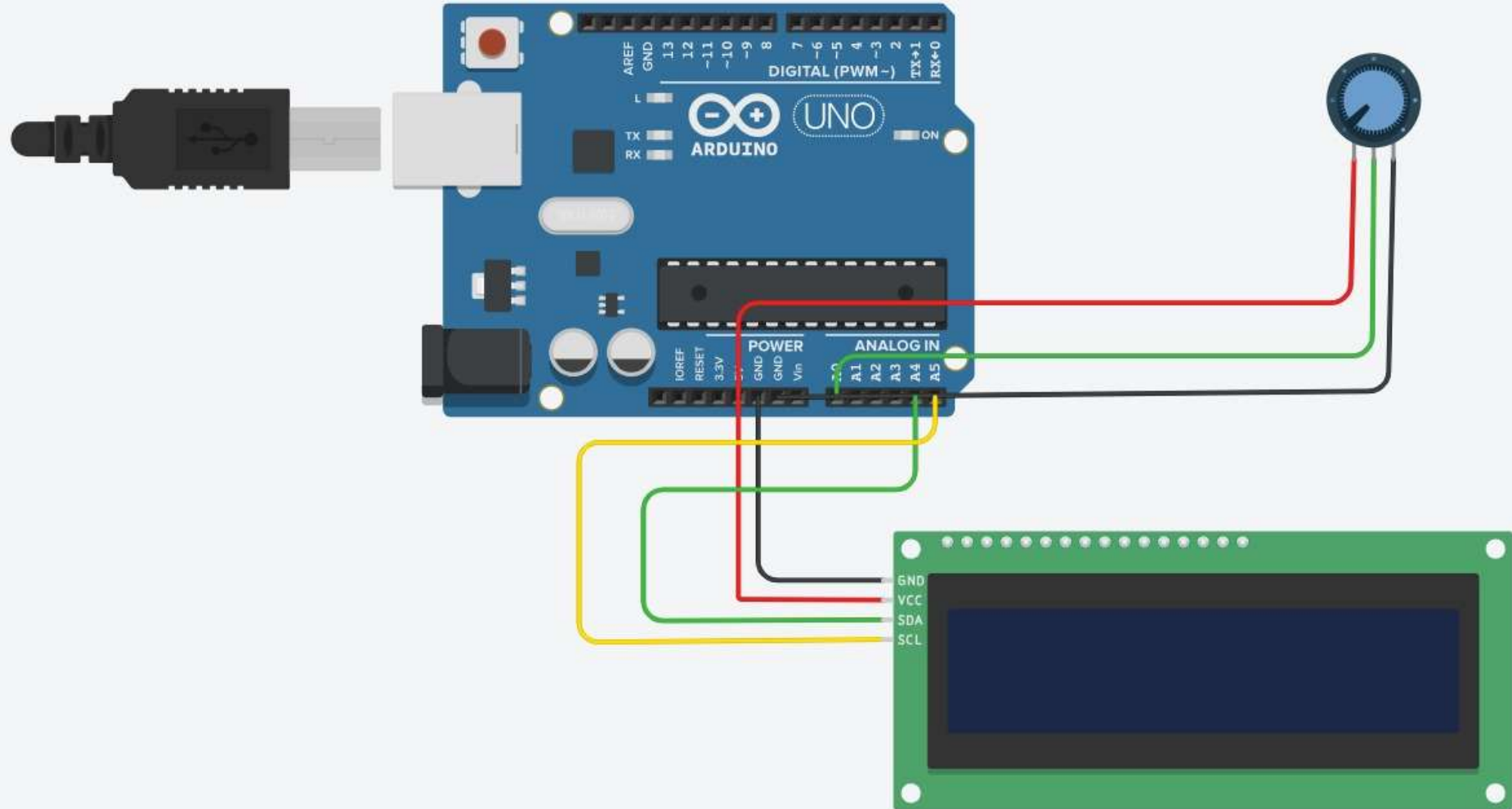
## Calibration Trimpot

Multi-turn cermet potentiometer for reference span tuning.

INTERNAL COMPONENTS

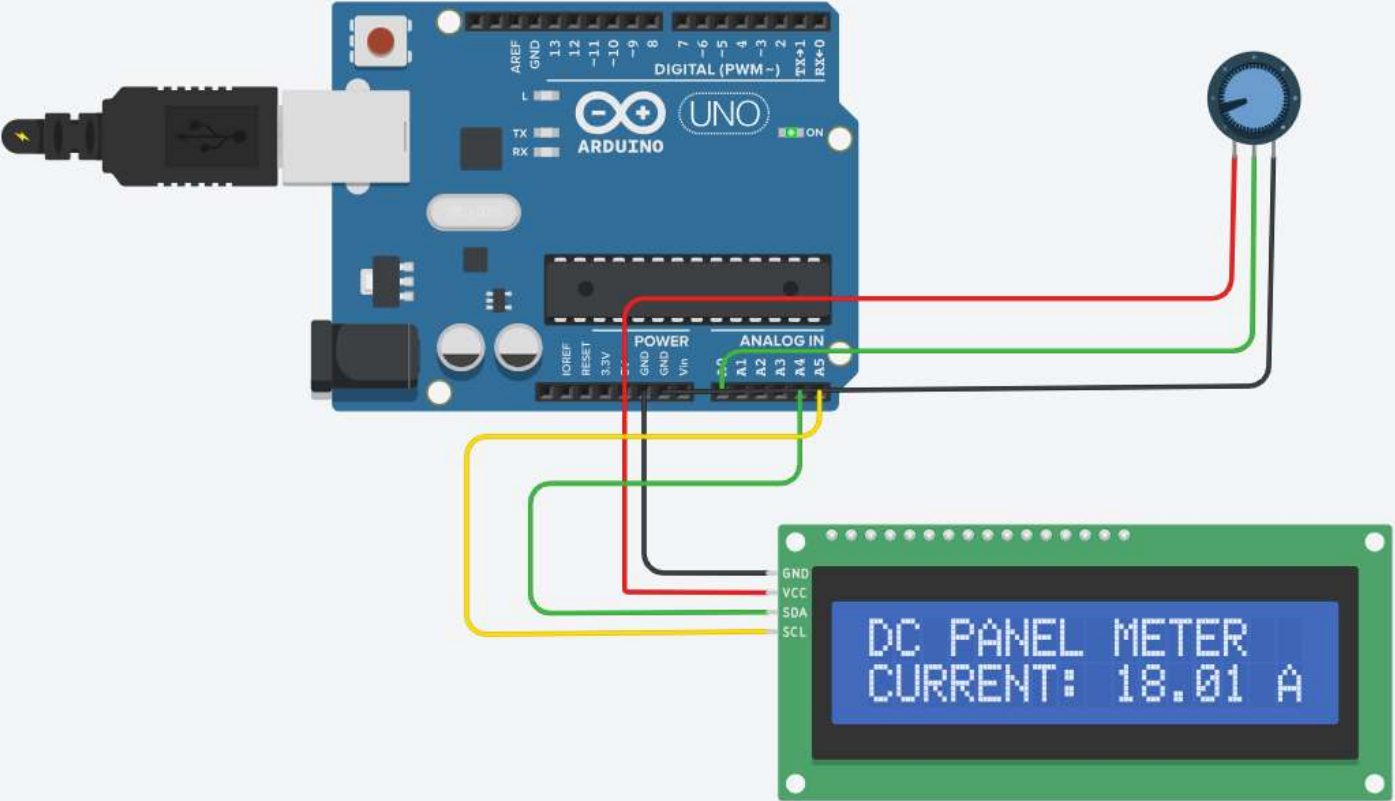
|                                                                                      |                                                                                       |                                                                                       |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|      |      |      |
| C1 electrolytic                                                                      | C2 electrolytic                                                                       | Diodes 1N4007                                                                         |
|    |    |    |
| Diodes D3                                                                            | Disc cap C9                                                                           | Disc caps C7 C8                                                                       |
|    |    |    |
| Display board                                                                        | Film cap 224J                                                                         | Film cap grey                                                                         |
|    |    |    |
| ICL7107 IC                                                                           | Orange electrolytic                                                                   | Power resistor                                                                        |
|    |    |    |
| Q1 TO92                                                                              | Resistor R1                                                                           | Resistor R10                                                                          |
|    |    |    |
| Resistor R11                                                                         | Resistor R5                                                                           | Resistor R6                                                                           |
|   |   |   |
| Resistor R9                                                                          | Seven segment displays                                                                | TO220 TR                                                                              |
|  |  |  |
| Terminal block                                                                       | Transformer                                                                           | VR1 trimmer                                                                           |
|  |                                                                                       |                                                                                       |
| Zener 5V1                                                                            |                                                                                       |                                                                                       |

# DC PANEL METER

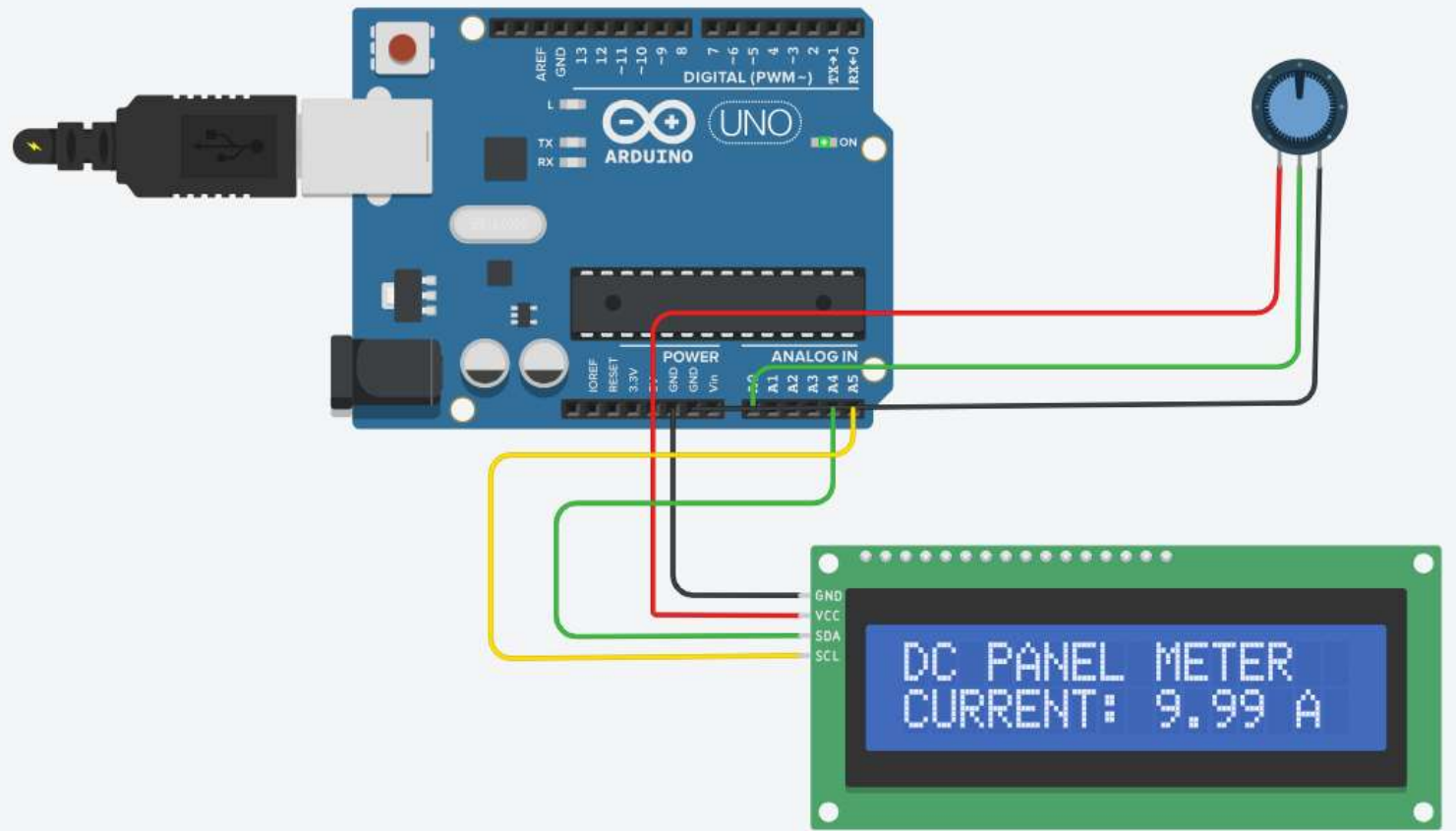


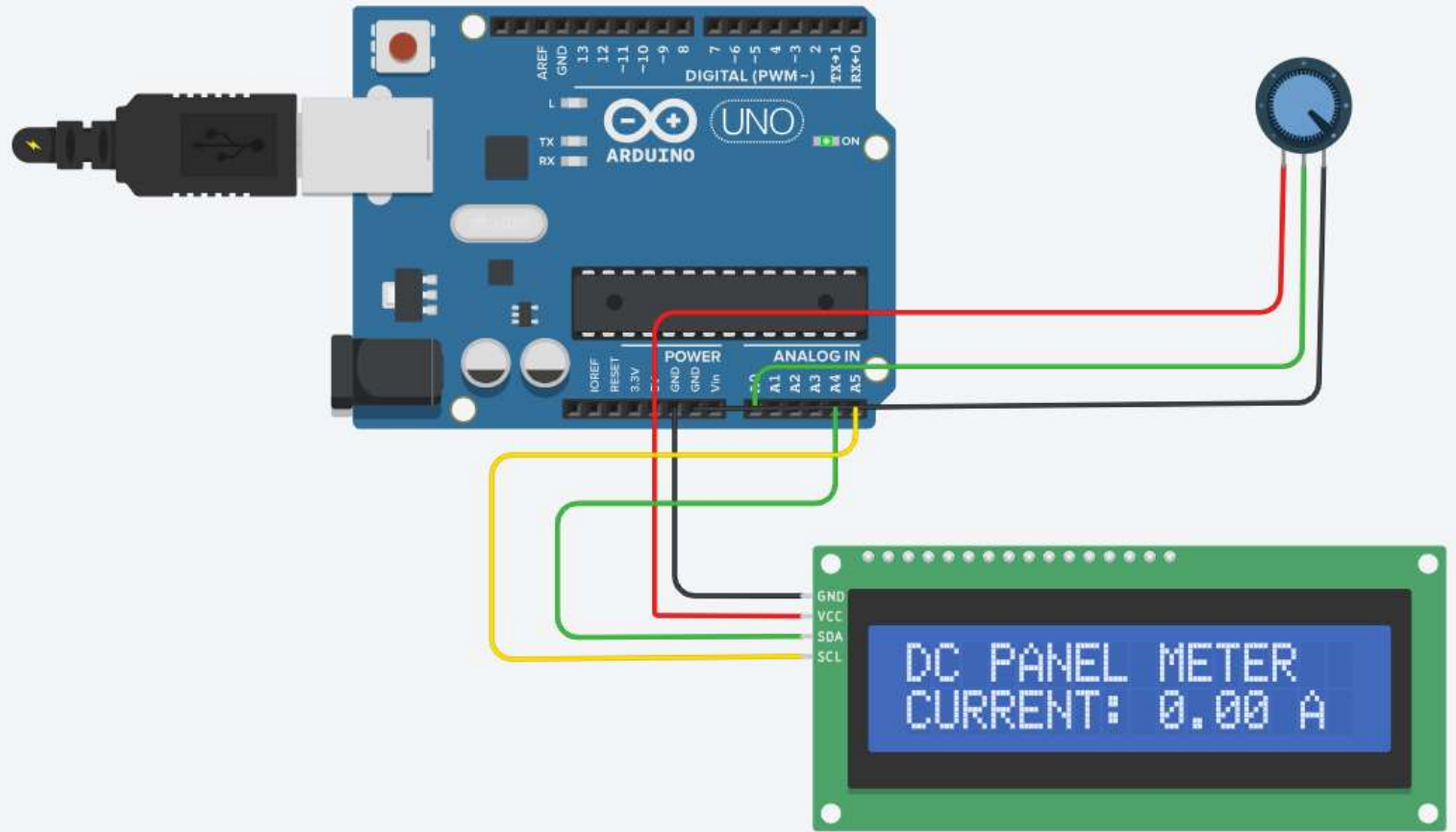
Simulator time: 00:00:13.150

Code Stop Simulation Send









DC PANEL METER SHOWS DIFFERENT READING ON CHANGING THE VALUE ON POTENTIOMETER

## ARDUINO CODE FOR DIGITAL AMMETER

```
1 #include <Adafruit_LiquidCrystal.h>
2
3 // This initializer specifically configures an MCP23008 chip at address 0 (whi
4 Adafruit_LiquidCrystal lcd(0);
5
6 const int analogInputPin = A0;
7 float maxSimulatedAmps = 20.0;
8
9 void setup() {
10     // Set up the MCP23008 LCD columns and rows
11     lcd.begin(16, 2);
12     lcd.setBacklight(HIGH);
13
14     // Meter boot-up sequence
15     lcd.setCursor(0, 0);
16     lcd.print(" M-TECH DIGITAL ");
17     lcd.setCursor(0, 1);
18     lcd.print("  AMMETER V1.0 ");
19     delay(2000);
20     lcd.clear();
21 }
22
23 void loop() {
24     // Read current dial value from pin A0 (0 to 1023)
25     int rawRead = analogRead(analogInputPin);
26
27     // Math mapping to convert the dial position into an Amperage range (0-20A)
28     float measuredCurrent = (rawRead / 1023.0) * maxSimulatedAmps;
29
30     // Render Panel Interface
31     lcd.setCursor(0, 0);
32     lcd.print("DC PANEL METER ");
33
34     lcd.setCursor(0, 1);
35     lcd.print("CURRENT: ");
36
37     // Display clean current reading matching your real panel meter
38     if (measuredCurrent < 0.1) {
39         lcd.print("0.00");
40     } else {
41         lcd.print(measuredCurrent, 2);
42     }
43
44     lcd.print(" A  ");
45
46     delay(300);
47 }
```



# Measurement & Power Signal Flow

## STAGE 01

### Current Shunt

Current drop converts input into millivolt DC signal.

## STAGE 02

### Dual-Slope ADC

ICL7107 executes auto-zero and dual-slope conversion.

## STAGE 03

### Direct Drive

Decoded 7-segment lines drive common anode displays.

## STAGE 04

### Linear Supply

Mains transformer steps down 220V to regulated rails.

REVERSE ENGINEERING EVALUATION

# Design Insights & Architecture Trade-offs

## Zero Firmware

Pure hardware ASIC design eliminates code maintenance and clock drift.

## Galvanic Isolation

Dedicated transformer isolates 220V mains noise from current sensing.

## Repairability

Standard through-hole components allow effortless diagnostic and field repair.

Q&A • Discussion & Analysis